

Experimental Analysis of a Building-Integrated Photovoltaic and thermal (BIPVT) Module with Detachable Phase Change Material (PCM) for Curtain Wall Applications

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Abstract. The increase in panel temperature in a photovoltaic thermal system (PVT) leads to a degradation of power generation performance, making it essential to suppress the module's temperature rise. This study developed a Building integrated photovoltaic thermal system (BIPVT-PCM) module with a phase change material (PCM) attached to the rear of a BIPVT module. This module has the advantage of a detachable PCM and can be easily installed on buildings in a curtain wall form. To ensure experimental accuracy, experiments were conducted in three phases during the winter. Phase 1 compared the power generation performance between Building integrated photovoltaic system (BIPV) modules, Phase 2 compared the thermal and power generation performance between BIPVT modules, and Phase 3 applied PCM to only one module to compare the performance of each module. The analysis results showed that the average electrical efficiency for both modules was 11.5%, with no significant performance improvement observed. The thermal efficiency decreased by 14.2% with the attachment of PCM. This was because although the heat generated during module operation was transferred to the PCM, the direct attachment method to the rear of the panel caused the transferred heat to re-influence the panel's temperature. However, thermal imaging analysis showed that the PCM prevented a rapid increase in the panel's temperature. This indicates that while the PCM contributes to the temperature stabilization of the panel, its latent heat storage properties caused a delay in the heat recovery time, making it disadvantageous for real-time heat recovery. Therefore, directly attaching PCM to the rear of the BIPVT module is not effective for improving the module's thermal performance. This suggests that when applying PCM to a BIPVT, a design for an alternative attachment method is necessary

Keywords. Building integrated photovoltaic thermal system. photovoltaic thermal systems. phase change material. (up to five keywords)

1 Introduction

In response to global climate change and the increasing need to achieve carbon neutrality, worldwide interest in renewable energy technologies has continued to grow [1]. In the building sector, numerous studies have examined ways to enhance energy self-sufficiency and self-consumption by integrating renewable energy systems into building [2]. Geothermal systems, heat pumps utilizing renewable sources, and solar thermal technologies have been widely explored to reduce the reliance on fossil fuels and promote the use of sustainable energy resources [3].

This study focuses on photovoltaic thermal (PVT) systems among various renewable energy technologies. Photovoltaic systems generate electricity from incident solar radiation; however, their electric performance is adversely affected by panel temperature rise [4]. Photovoltaic-thermal (PVT) systems address this

limitation by simultaneously removing excess heat from the PV panel and converting it into useful thermal energy, thereby enabling co-generation of electricity and heat [5-6]. Conventional PVT systems are typically installed on building rooftops or in open open-land areas where sufficient space is available.

However, when the PVT module is mounted on the building façade—as in a building-integrated photovoltaic/thermal (BIPVT) system—it can significantly enhance the energy self-sufficiency of multi-family residential buildings with limited rooftop area [7]. Because the system is integrated into the exterior wall, the electricity generated by the PVT module can be used according to the building's demand, while the collected heat can be supplied for space heating and domestic hot water.

In particular, this study was conducted under winter outdoor conditions from a short-term performance

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perspective to verify the applicability of a detachable PCM integrated into a BIPVT system. By attaching PCM to the rear side of the BIPVT module, the latent heat behavior of the PCM under this attachment configuration was analyzed based on the time-dependent thermal collection, panel surface temperature, and electrical power output of the BIPVT-PCM system. A new BIPVT module incorporating phase change material (PCM)—referred to as a BIPVT-PCM system—was fabricated, and its performance was experimentally evaluated. To further enhance the performance of the BIPVT system, the proposed module employs PCM to stabilize the operating temperature of the panel and to store the recovered thermal energy.

2 Methodology

2.1 Overview of BPVT-PCM System

The BIPVT module used in this study adopts a vertical, frame-free curtain-wall configuration that allows it to function as part of the building façade. This configuration integrates the functions of a photovoltaic-thermal system and a building envelope component. Fig. 1(a) illustrates the BIPVT module, while Fig. 1(b) presents the BIPVT-PCM module with PCM attached.

Table 1. Properties of the paraffin wax PCM

Property of paraffin wax PCM	Values	Units
Melting temperature range	29 - 36	°C
Melting Onset Temperature	30.3	°C
Melting Peak Temperature	35.8	°C
Latent Heat of Fusion	184.9	J/g
Package Size	230×180×25	mm
Attachment Method	Silicone, Thermally conductive adhesive tape	-
Quantity	15	-
Analysis Conditions	Values	Units
Analysis Equipment	DSC 250 (TA instruments)	-
Heating Rate	1	°C/min
Maximum Temperature	100	°C
Atmosphere Gas	N2	-

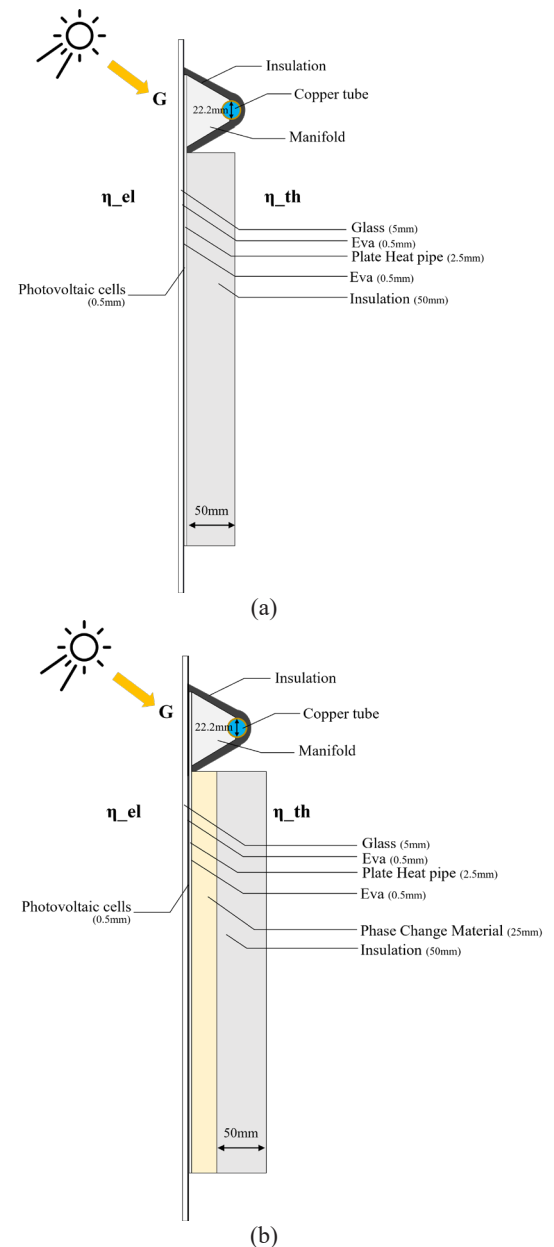


Fig. 1. Sectional view of the BIPVT panel: (a) BIPVT; (b) BIPVT-PCM.

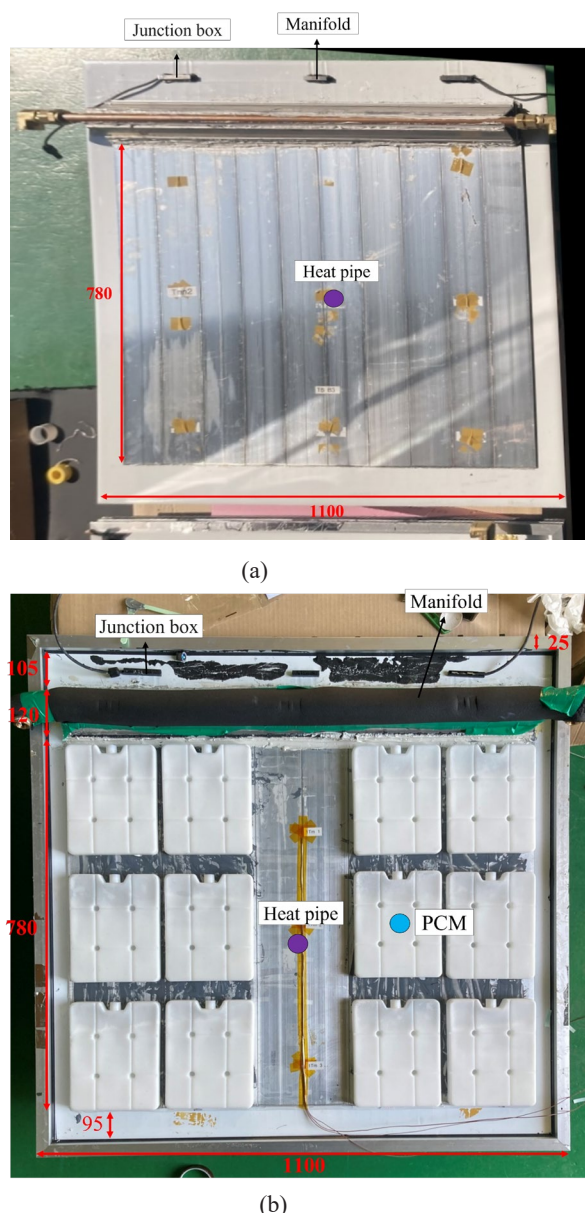


Fig. 2. Prototype panels of BIPVT and BIPVT-PCM: (a) rear view of prototype BIPVT panels; (b) rear view of prototype BIPVT-PCM panels.

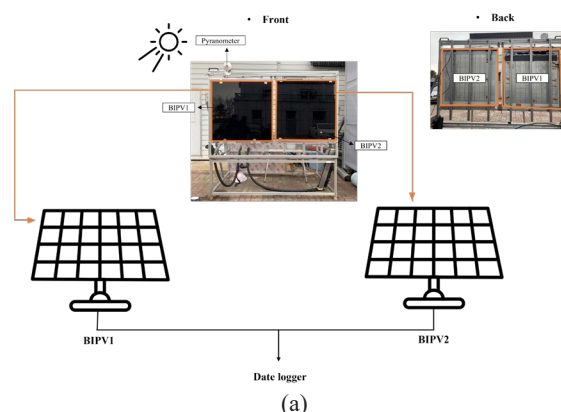
Electricity generation is achieved through an inverter mounted on the rear side of the module. For thermal collection, the integrated microchannel flat-plate heat pipe (MFHP) transports heat from the lower section to the upper section of the module, where it is recovered through a manifold. Plastic-pack PCM units were installed on top of the MFHP to construct the BIPVT-PCM module, as illustrated in Fig. 2. The PCM used in this study is paraffin-wax-based, and its detailed properties are summarized in Table 1.

2.2 Experimental Setup and Conditions

Table 2 Summary of test condition

Test	Test name	Test period	Module configuration	
			Module (1)	Module (2)
Test 1	BIPV test	2025-1-4 ~ 2025-1-17	BIPV (1)	BIPV (2)
Test 2	BIPVT test	2025-2-3 ~ 2025-2-16	BIPVT (1)	BIPVT (2)
Test 3	BIPVT-PCM test	2025-2-21 ~ 2025-3-9	BIPVT (1)-PCM (with PCM)	BIPVT (2) (without PCM)

The experiments were conducted outdoor in the Plus Energy Community at the Korea Institute of Energy Research (KIER) located in Daejeon, South Korea. The study consisted of three sequential test stages. The overall experimental system configuration and procedures are presented in Fig. 3 and Table 2. In Test 1, two identical BIPV modules were evaluated under the same environmental conditions to compare their inherent power generation performance. This step was conducted to verify that no inherent performance deviation existed between the two panels prior to the main experimental evaluation. In Test 2, a manifold was attached to the rear side of each module to configure BIPVT modules, and their electrical generation and thermal collection performance were examined. In other words, the BIPV panels used in Test 1 were retrofitted with manifolds to fabricate BIPVT panels, and their performance was evaluated to confirm that no significant performance difference existed between the two panels even after the thermal collection system was added. Finally, in Test 3, PCM was applied only to the rear side of Module (1), and the performance of the two modules was compared to investigate the effect of PCM integration. Through this three-stage experimental process, external variables were minimized, enabling accurate and reliable performance evaluation.



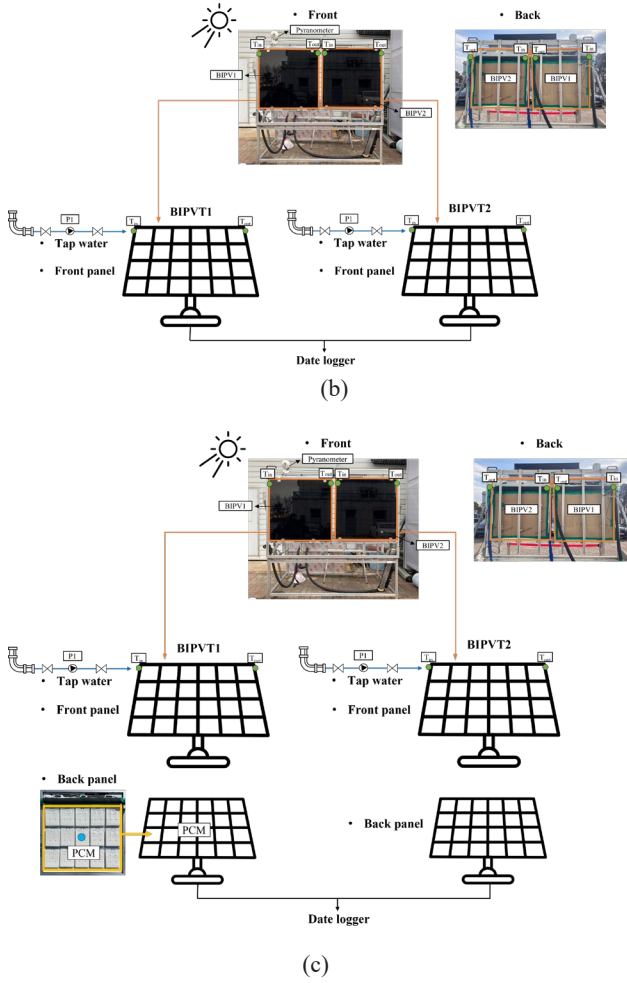


Fig. 3. overall system configuration: (a) Prototype BIPV panels for Test 1; (b) Prototype BIPV panels for Test 2; (c) Prototype BIPV panels for Test 3.

The governing equations are as follows. The collected thermal energy is denoted as Q (W). The volumetric flow rate is represented by V (m^3/s). The specific heat capacity, C_p is taken as $3.75 \text{ kJ/kg } ^\circ\text{C}$, and ρ is the fluid density, assumed to be 1000 kg/m^3 . The inlet and outlet temperatures, T_{in} and T_{out} , are expressed in degrees Celsius ($^\circ\text{C}$).

$$Q = V \times C_p \times \rho \times (T_{out} - T_{in}) \quad (1)$$

The solar irradiance, $G(\text{kW})$, is defined in Eq. (2). $I(\text{W/m}^2)$ represents the incident solar radiation per unit area, and A is the surface area of the BIPVT panel, which is 1.21 m^2 .

$$G = \frac{I \times A}{1000} \quad (2)$$

The thermal collection efficiency (η_Q) and electrical efficiency (η_p) are calculated using Eq. (3) and Eq. (4), respectively.

$$\eta_Q = \frac{Q}{G} \times 100 \quad (3)$$

$$\eta_p = \frac{p}{G} \times 100 \quad (4)$$

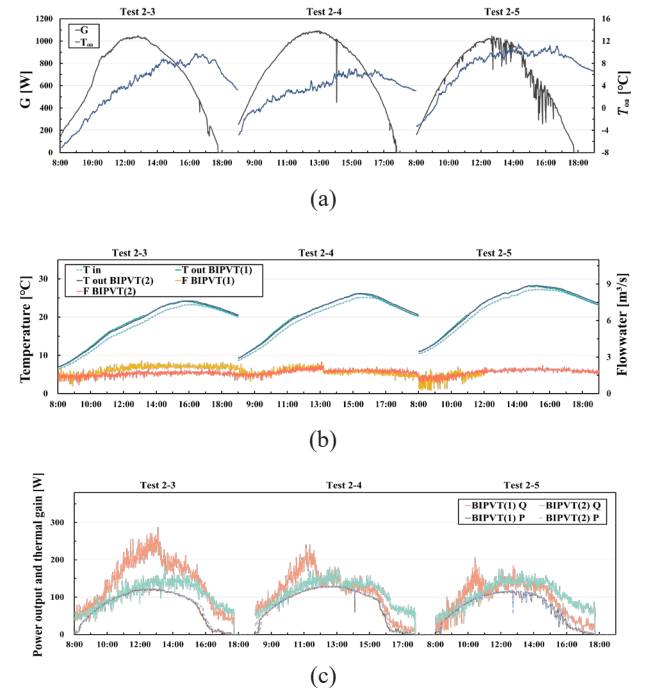
3 Results

3.1 Verification of BIPVT Modules: Test 1 and 2

3.1.1 Test 1: Electrical Performances of the Frameless BIPVT Modules

Test 1 compared the performance of the two modules under identical conditions to ensure the reliability of subsequent tests. The average electrical efficiency of Module (1) was 11.4%, while that of Module (2) was 11.7%. Although Module (1) showed a 0.3%p lower efficiency, this difference can be attributed to natural variations in module performance over the extended testing period. Therefore, the two modules can be considered to have essentially similar performance characteristics.

3.1.2 Electrical and Thermal Performances of Frameless BPVT Modules



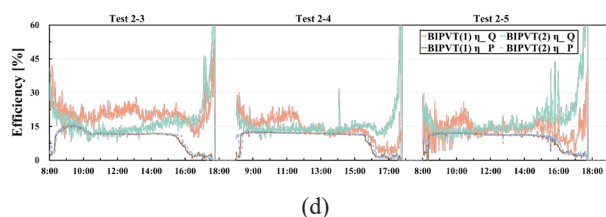


Fig. 4. Comparison of power and thermal performances of the modules in Test 2: (a) environmental conditions; (b) inlet and outlet temperature and flow rate; (c) power output and thermal gain; (d) electrical and thermal efficiencies

Test 2 compared the electrical and thermal performance of the two modules. The performance curves for each experimental case in Test 2 are presented in Fig. 4.

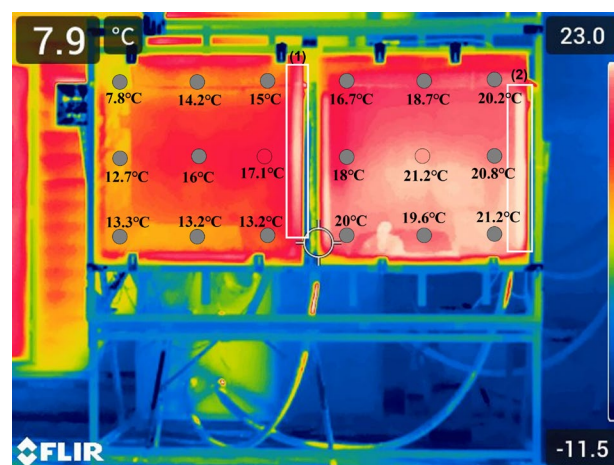
The average electrical efficiencies were 11.4% for Module (1) and 11.7% for Module (2), resulting in a 0.3% difference between the two. Considering that Test 1 also showed a 0.3%p difference, it can be interpreted that both modules continued to exhibit comparable electrical performance in Test 2. In terms of thermal performance, the average thermal efficiency of Module (1) was measured as 15.7%, whereas Module (2) recorded 13.3%. Thus, Module (1) demonstrated approximately 17.5% higher thermal efficiency (a 2.4 %p) compared to Module (2).

Overall, the results from Test 1 and Test 2 indicate that the two modules exhibit similar electrical performance, while Module (1) consistently achieves notably higher thermal performance—approximately 17.5% greater than that of Module (2).

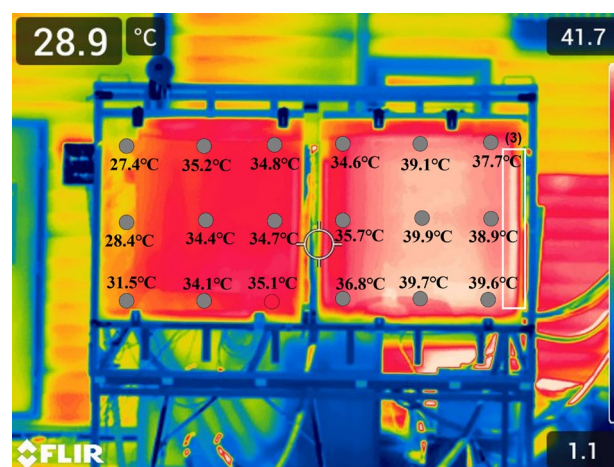
3.2 Performance Comparison of BIPVT and BIPVT-PCM: Test 3

Test 3 was conducted by attaching PCM to Module (1) to construct the BIPVT-PCM module and comparing its electrical and thermal performance with that of the BIPVT Module (2), which did not contain PCM.

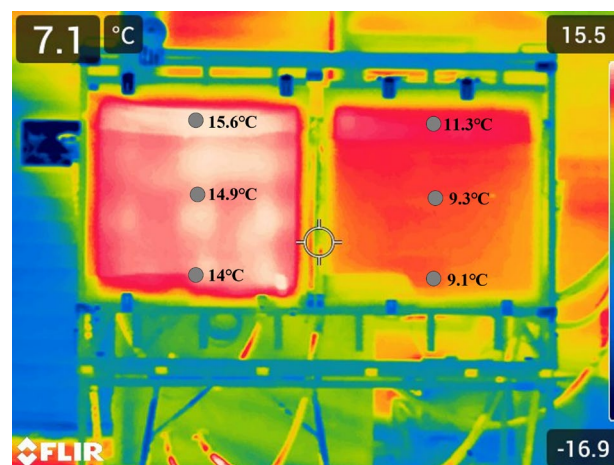
In terms of electrical performance, both Module (1) and Module (2) exhibited an identical average electrical efficiency of 11.5%. Compared with the previous tests, the efficiency of Module (1) increased from 11.4% to 11.5%, which can be attributed to the PCM mitigating rapid temperature rise in the module and thereby slightly enhancing its power-generation performance. For thermal performance, the average thermal efficiency of Module (1) was 10%, whereas Module (2) recorded 11.7%. Despite the integration of PCM, Module (1) showed a 14.2% (1.7 %p) reduction in thermal efficiency relative to Module (2). The detailed results for Tests 1–3 are summarized in Table 3. Figure 5 and Table 4 present the time-dependent surface temperature distribution of the two modules obtained through thermal imaging analysis during Test 3.



(a)



(b)



(c)

Fig. 5. Comparison of surface temperatures of BIPVT(1)-PCM and BIPVT(2) modules

Table 3 Overview of Tests

Tests	T_{OA} [°C]	Total G [kWh]	T_{in} [°C]	T_{out} [°C]		V [m³/s]		Total Thermal gain		Total Power output		Efficiency			
								Q [kWh]		P [Wh]		η_Q [%]		η_P [%]	
								(1)	(2)	(1)	(2)	(1)	(2)	(1)	(2)
Test 1	1.0	24.9	-	-	-	-	-	-	-	2.8	2.9	-	-	11.4	11.7
Test 2	2.4	26.5	18.7	19.9	19.8	1.8	1.8	4.2	3.5	3	3	15.7	13.3	11.4	11.7
Test 3	5.8	31.1	23.7	24.5	24.7	1.6	1.6	3.1	3.6	3.5	3.6	10.0	11.7	11.5	11.5

Table 4 Overview of BIPVT(1)-PCM and BIPVT(2) modules

Tests	T_{OA} [°C]	Total G [Wh]	T_{in} [°C]	T_{out} [°C]		V [m³/s]		Total Thermal gain		Total Power output		Efficiency			
								Q [Wh]		P [Wh]		η_Q [%]		η_P [%]	
								(1)	(2)	(1)	(2)	(1)	(2)	(1)	(2)
8:00	-4.3	356.8	8.1	8.1	8.0	0.0	0.1	0.0	0.3	48.4	47.9	0.0	0.1	17.6	17.4
9:00 (Fig. 3(a))	-1.5	655.4	13.4	14.2	14.1	0.9	0.9	49.5	44.8	82.7	81.9	7.0	6.3	12.6	12.5
10:00	-0.7	845.2	16.3	17.4	17.2	1.9	1.8	123.6	96.2	106.8	106.2	14.6	11.4	12.6	12.6
11:00	0.8	964.3	19.1	20.1	20.1	2.0	1.9	131.1	116.8	119.2	117.9	13.6	12.1	12.4	12.2
12:00	2.6	1023.7	21.5	22.3	22.5	2.2	1.9	113.6	126.5	123.5	122.8	11.1	12.4	12.1	12.0
13:00	3.6	1012.9	23.1	23.9	24.1	2.2	2.0	114.7	131.4	121.0	120.4	11.3	13.0	11.9	11.9
14:00 (Fig. 3(b))	4.5	934.9	24.3	25.2	25.4	2.0	1.9	103.4	120.8	109.5	109.1	11.0	12.9	11.7	11.7
15:00	5.1	795.1	25.0	25.6	25.9	1.7	1.8	72.4	100.6	75.7	85.0	9.1	12.6	9.4	10.6
16:00	4.7	593.0	24.1	24.4	24.7	1.6	1.7	26.5	59.6	12.7	12.9	4.4	10.0	2.1	2.1
17:00 (Fig. 3(c))	4.0	310.3	22.1	22.2	22.5	1.5	1.7	8.1	40.9	3.9	4.4	2.1	14.4	1.0	1.2
18:00	2.2	0.0	20.0	20.2	20.3	1.2	1.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

The BIPVT(1)-PCM module exhibited higher thermal performance than BIPVT(2) during the morning period (09:00–12:00), with a maximum thermal efficiency of 14.6%. However, in the afternoon period when solar irradiance reached its peak (after 12:00), its maximum thermal efficiency decreased to 11.3%, falling below that of BIPVT(2), which reached 14.4%. This behavior can be explained by the latent heat storage characteristics of the PCM: the stored heat contributes to higher thermal output during the morning hours, while after noon, when the BIPVT system becomes fully active, the PCM is unable to transfer the newly collected heat in real time, resulting in reduced thermal efficiency.

Thermal imaging analysis further supports this interpretation. At 14:30, when solar irradiance was at its maximum, the BIPVT(1)-PCM module reached a peak surface temperature of 35.2 °C, which was lower than the 40.5 °C observed for the BIPVT(2) module. This indicates that the PCM attenuates rapid temperature increases by absorbing heat. During nighttime, the PCM continued to retain stored heat, causing the BIPVT(1)-PCM module to maintain a higher surface temperature than BIPVT(2). This suggests that the PCM contributes to stabilizing panel temperature by preventing rapid temperature drops during non-irradiated hours.

3.3 Discussion

This study fabricated a BIPVT-PCM module by integrating PCM onto the rear side of a BIPVT panel and conducted a comparative analysis of its electrical and thermal performance. When PCM was applied, the electrical performance remained stable, unlike the trends reported in previous studies; however, the thermal performance decreased by 14.2%. This reduction can be attributed to the latent heat characteristics of the PCM. When heat is generated in the BIPVT-PCM module, the PCM absorbs the heat during its phase-change process rather than allowing it to be immediately transferred to the working fluid, resulting in lower heat recovery at the manifold.

In addition, the specific design configuration—attaching plastic-pack PCM directly on top of the rear MFHP—is interpreted as a major factor that hindered thermal performance enhancement. In the original BIPVT modules used in Test 2, heat is transported from the lower to the upper section of the heat pipe and recovered through the manifold into the working fluid. However, in the BIPVT-PCM module, the heat that would normally migrate through the heat pipe is first absorbed and stored by the PCM, and this stored heat is not transferred promptly to the manifold, leading to a reduction in thermal output. To ensure a valid and clear evaluation of the system behavior, this study cross-validated the variation in the front surface temperature of the BIPVT module with the corresponding electrical power output and thermal gain at each time interval, rather than relying on direct PCM surface temperature measurements.

Although the curtain-wall configuration provides practical advantages by enabling easy installation on building façades, the current PCM attachment strategy delays real-time heat transfer and consequently results in lower thermal performance.

4 Conclusion

4.1 Conclusion and Future Work

This study integrated PCM into a frameless BIPVT module and compared its electrical and thermal performance under outdoor winter conditions. The results for each test are presented in Fig. 6.

In terms of electrical performance, Module (1) and Module (2) exhibited efficiencies of 11.4% and 11.7%, respectively. For thermal performance, the BIPVT modules without PCM achieved thermal efficiencies of 15.7% for Module (1) and 13.3% for Module (2).

Subsequently, Test 3 evaluated the performance after PCM integration. The electrical efficiency of both modules increased slightly to 11.5% following PCM application. However, for thermal performance, the PCM-applied Module (1) showed a 14.2% reduction in thermal efficiency compared with Module (2).

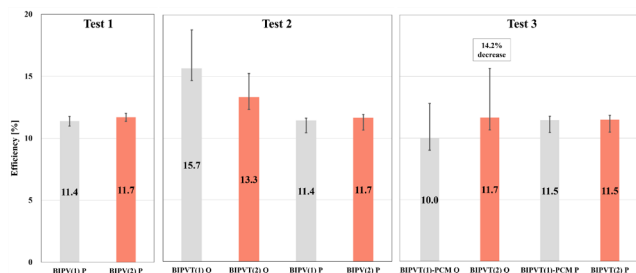


Fig. 6 Summary of Test 1 – 3

As shown in Table 5, recent studies have predominantly applied PCM to air-type or water-type PVT systems installed at tilted angles. In contrast, the present study examined a frameless, vertically installable curtain-wall BIPVT-PCM configuration. This design enables the application of renewable energy even in buildings with limited rooftop area.

The findings also provide a new experimental insight, demonstrating that directly attaching PCM to the rear side of the panel can lead to a reduction in thermal performance. This highlights an important academic implication regarding the limitations of the current attachment method. Therefore, future research should focus on developing an improved PCM integration strategy. Moreover, to ensure both ease of installation and stable electrical and thermal performance in BIPVT-PCM systems, optimization of the module configuration is essential. Further investigations into the PCM phase-change temperature, attachment configuration, and heat-transfer pathway are expected to contribute to enhanced performance of building-integrated photovoltaic-thermal modules.

Table 5 Comparison of BIPVT-PCM performance with previous studies

Reference	PVT type	PCM type/Melting point	Test condition	Avg. electrical efficiency	Avg. thermal efficiency
Preet et al.(2017) [8]	Flat plate PVT	Paraffin wax RT /30 °C	Outdoor	+70% vs. PVT	-26.97% vs. PVT
Kazemian et al.(2018) [9]	Flat plate PVT	Organic paraffin wax	Outdoor	+8.81% vs. PV (exergy)	1.04%
Chavan and Devaprakasm (2019) [10]	Flat plate PVT	white petroleum jelly PCM /36–60 °C	Outdoor	15.32% (+4.24% vs. PV)	45.75%
Maatallah et al. (2019) [11]	Flat plate PVT	Paraffin wax	Outdoor	+17.33% vs. PV	26.87%
T Maatallah et al. (2019) [11]	Flat plate PVT	Paraffin/57 °C	Outdoor	+17.33% vs. PV	+26.87% vs. PV
Modjinou et al. (2019) [12]	Flat plate PVT	Macro-encapsulated PCM	Outdoor	+4.86%p vs. PVT	+4.93%p vs. PVT
Carmona et al. (2021) [13]	Flat plate PVT	RT-35/29–36 °C	Outdoor	+7.43% vs. PV	31.35% (Overall eff. PVT-PCM)
This study	BIPVT-PCM	Paraffin Wax / 29–36°C	Outdoor, vertical (winter season)	-	-14.2% vs. BIPVT

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